

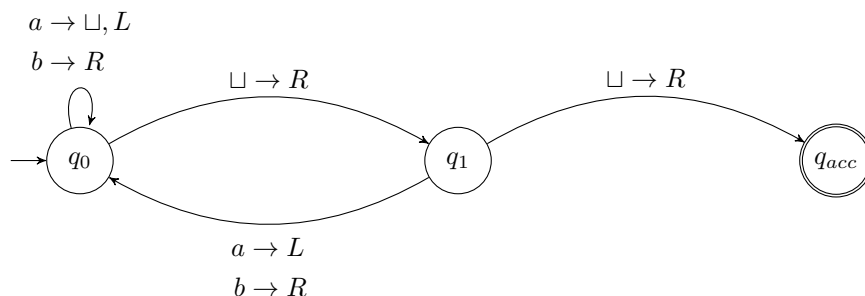
CSE2315 Lab Course

Solutions 4

deadline: March 6, 2025, 13:45

1. *Old exam question, endterm 2022-2023:* Given the Turing Machine M below. You may assume the tape is bounded on the left.

You may assume that any missing transitions lead to the reject state, which has been left out.



- (a) Give two words that are in $L(M)$ and two words that are not. For at least one of each, explain why not by giving the sequence of configurations the machine goes through. All words should be at least length 2.

Solution: For example:

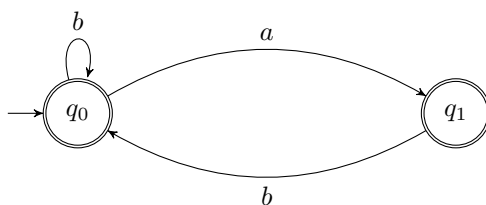
- $ab \in L(M)$ as $q_0ab\sqcup \vdash q_0\sqcup b\sqcup \vdash \sqcup q_1b\sqcup \vdash \sqcup bq_0\sqcup \vdash \sqcup b\sqcup c_1\vdash \sqcup b\sqcup \sqcup q_{acc}$
- $aa \notin L(M)$

- (b) Describe the language $L(M)$ (you can do this informally or in set-builder notation).

Solution: The language of words that do not have 2 a 's directly following each other.

- (c) Is $L(M)$ decidable? Explain your answer.

Solution: Yes. $L(M)$ is described with the following NFA:



Languages that can be described with an NFA are regular, and thus also decidable.

2. Suppose we have the following language over the alphabet $\Sigma = \{1, \dots, 9, \#\}$:

$$L = \{v\#w \mid v, w \in \{1, \dots, 9\}^* \wedge \exists a, b \in \{1, \dots, 9\} : c_a(v) = c_b(w)\},$$

where $c_x(y)$ denotes the number of occurrences of symbol x in word y . Give a high-level description of a nondeterministic Turing machine that decides L .

Solution: A high-level description for a machine M deciding L is given by:

$M =$ "On input $w \in \Sigma^*$:

1. Check if there is exactly one $\#$ symbol in w . If not, reject.
2. Nondeterministically decide on symbols a and b from $\{1, \dots, 9\}$.
3. Go to the first character on the tape.

4. Check the current character on the tape.
 - a. If it is #, go right until the first unmarked b (if any). If there is such a b , reject. Else, accept.
 - b. Else if it is a , mark it and go to step 5.
 - c. Else, mark it, go right and go to step 4.
5. Go right until the first character after the # symbol.
6. Check the current character on the tape.
 - a. If it is marked, go right and go to step 6.
 - b. If it is b , mark it and go to step 7.
 - c. If it is $\sqcup$, reject.
 - d. Else, mark it, go right and go to step 6.
7. Go left to the first unmarked character on the tape and go to step 4.

3. Suppose we have the following language over the alphabet $\Sigma = \{0, 1, \#\}$:

$$L = \{0^n \# 1^m \mid n > m\}$$

Describe an order in which we can enumerate L . Does your order show that L is decidable?

Solution: First do all words of length 2 ($0\#$), then all words of length 3 ($00\#$), then length 4 ($000\#, 00\#1$), etc. Sort within each group by the number of zeroes at the start.

This proves L is decidable as for a given input we can run the enumerator and then simply reject when we get to a word longer than the input, or accept if we find it.

4. Consider the following language:

$$L = \{\langle v_1, v_2, \dots, v_k, D \rangle \mid v_1, v_2, \dots, v_k \in \{0, 1\}^* \text{ and } D \text{ is a DFA with } L(D) = \overline{K}, \text{ where } K = \{v_1, v_2, \dots, v_k\}\}.$$

Is the following Turing machine M a decider for L ? Explain.

$M =$ "On input $w = \langle v_1, v_2, \dots, v_j, D \rangle$:

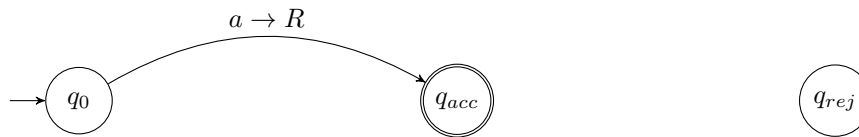
1. For $i = 1, \dots, j$:
 - a. Run D on v_i .
 - b. If D accepts, reject.
2. If D has rejected all v_i , accept."

Solution: No, M does not decide L . It only checks whether $\{v_1, \dots, v_j\}$ is a subset of $\overline{L(D)}$. For example, M always accepts if $L(D) = \emptyset$. Alternatively: Take a DFA D' such that $L(D') = \{0, 01\}$. As an input to M take $\langle 11, D' \rangle$. M accepts this input, but $L(D') \neq \overline{\{11\}}$, so it should reject this input. Therefore M is not a decider for this language.

5. Old exam question, resit 22-23:

- (a) (1 point) Given a Turing Machine M with input alphabet Σ and tape alphabet Γ . Is it possible for $|\Gamma| = |\Sigma| + 2$? If yes, draw a valid TM M for which this holds. If not, explain why not.

Solution: Yes, the following machine has $\Sigma = \{a\}$ and $\Gamma = \{a, \sqcup, x\}$. All missing transitions lead to q_{rej} .



Remember that Γ must always have $\sqcup$ in it as well! Any Γ without it, is not valid for a Turing Machine.

Source: Sipser exercise 3.5b

- (b) (1 point) Can a Turing Machine contain just a single state? If yes, draw a valid Turing Machine with a single state. If not, explain why not.

Solution: No, because a Turing Machine needs to have an accept state and a reject state, and they cannot be the same state.

Source: Sipser exercise 3.5d

6. *Revision, old exam question midterm 18-19:* Consider the following language, where the alphabet $\Sigma = \{a, b\}$:

$$L = \{b^n abv \mid v \in \Sigma^*, n \geq 1, \text{ and } |v| = 8n + 4\}$$

This language is **not** regular. Prove this using the pumping lemma in at most 20 lines.

Solution:

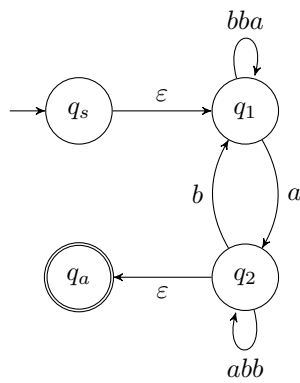
Proof. Proof by contradiction:

- Suppose L is regular.
- This means there must exist some pumping length $p > 0$ for L such that all words w longer than p can be split up into three parts x, y and z , with $|y| > 0$ and $|xy| \leq p$.
- For this division of w , and any $i \geq 0$, $xy^i z \in L$.
- Let's take the word $w = b^p aba^{8p+4}$ which is in L .
- This word is longer than p , so the above holds for this word.
- Given the requirements, we know that there is only one possible split
 - $x = b^\alpha, y = b^\beta$ and $z = b^{p-\alpha-\beta} aba^{8p+4}$, with $0 \leq \alpha < p, 0 < \beta \leq p$ and $\alpha + \beta \leq p$.
 - * Now, taking $i = 0$, we get $b^\alpha b^{p-\alpha-\beta} aba^{8p+4} = b^{p-\beta} aba^{8p+4}$, which is clearly not in L since $\beta > 0$.
- We have obtained a contradiction, so L must not be regular.

□

7. *Revision, variation on old exam question midterm 20-21:*

Given the GNFA G depicted below. If we rip q_2 out of G using the procedure described in Sipser, what (non-empty) transitions would be left in G . Make sure to describe all transitions.

**Solution:**

- $\delta(q_s, q_1) = \varepsilon$
- $\delta(q_1, q_1) = bba \cup a(abb)^*b$
- $\delta(q_1, q_a) = a(abb)^*$

These are all transitions.

See next page for an optional extra exercise puzzle.

Optional extra exercise This last exercise is not part of the learning material. However, if you like puzzles and have some time left, you might like to do this challenging puzzle.

Another week, another mystical mansion. This mansion seems a bit more similar to the mansion from week 1. Teleportation devices cannot teleport to multiple rooms. However, they can still teleport to 0 rooms. Nothing else is known about the mansion. This is largely because the walls block radio-signals you send or receive. Luckily, because the maximum amount of rooms that fit the mansion is still 4 and you have already made a lot of robots to use, you can just try to brute-force your way through. You can now pre-program the robot to do several instructions. If the robot isn't outside at the end of the instruction list, you let it self-destruct. Normally, if there is 4 rooms, there would always exist at least 1 order of 4 instructions that leads outside.

First, you try $Left \rightarrow Left \rightarrow Left \rightarrow Left$. When you teleport the robot inside, you can hear that the robot uses a teleportation device inside. Then, it is silent. That means that the first *Left* instruction already lead to a teleportation device that teleports to 0 rooms. Another try, $Right \rightarrow Left \rightarrow Left \rightarrow Left$. This time, you hear the robot use 2 teleportation devices. Apparently, now the second *Left* lead to 0 rooms. Since *Right* seemed to do a bit better, let's immediately try $Right \rightarrow Right \rightarrow Right \rightarrow Left$. You hear the robot take 4 teleportation devices. Then, $Right \rightarrow Right \rightarrow Right \rightarrow Right$, which results in 4 teleportation devices, and then you hear the robot self-destructing. Next try, $Right \rightarrow Right \rightarrow Left \rightarrow Left$. You count the amount of teleportations devices used, and count to 6. That's strange, you gave only 4 instructions. Why didn't it already self-destruct? Let's try the last option that can be it, $Right \rightarrow Right \rightarrow Left \rightarrow Right$. Again, you count 6 teleportation devices. We are missing some information.

You suspect the teleportation devices can alter whether the robot goes to the next or the previous instruction. Also, you suspect the teleportation devices might be able to change the instruction that was just read. Given all this information, you can't be sure if there is a way to get outside. But which instructions can you try for which it is at least possible that it gets the robot outside?

Start	Left	Left	Left	Left	First teleportation device removes it
Start	Right	Left	Left	Left	Second teleportation device removes it
Start	Right	Right	Right	Left	Fourth teleportation device removes it
Start	Right	Right	Right	Right	Goes through four teleportation devices, then self-destructs
Start	Right	Right	Right	Left	Sixth teleportation device removes it
Start	Right	Right	Right	Right	Sixth teleportation device removes it

- Each room has 2 teleportation devices, a left one and a right one
- Teleportation devices are deterministic, the same device in the same room will always lead to the same room (or no room), changes the last read instruction to the same instruction, and either always lets the robot read the next instruction or always lets it read the previous instruction
- At least 1 teleportation device leads outside
- There are at most 4 rooms
- If the robot should read the previous instruction but there is none, it reads the current instruction again
- If the robot should read the next instruction but there is none, it self-destructs
- Different robots go in with instruction lists as given in the table with as an outcome the right side of the table, what is possibly a way to get the robot outside to collect the treasure?

Solution: $Right \rightarrow Right \rightarrow Right \rightarrow Right \rightarrow Right \rightarrow Left$